

1. Random variables

Definition 1.1 (Expectation). The expectation of a random variable X on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ is

$$\mathbb{E}[X] = \int X \, d\mathbb{P},$$

whenever well-defined.

Remark. We require at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ to be finite.

Proposition 1.2. For any $A \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}_A(X)].$$

Theorem 1.3 (Markov). Let $X \geq 0$. For all $a > 0$,

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}.$$

Corollary 1.3.1 (Chebyshev). Let $\mathbb{E}[X] = \mu$, $\text{var}[X] = \sigma^2$. Then,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

A. Appendix

Proof of Theorem 1.3.

$$\begin{aligned} \mathbb{P}(X > a) &= \mathbb{E}[\mathbf{1}_{(a, \infty)}(X)] && \text{(Proposition 1.2)} \\ &\leq \mathbb{E}\left[\left(\frac{X}{a}\right) \mathbf{1}_{(a, \infty)}(X)\right] \\ &\leq \frac{\mathbb{E}[X]}{a}. \end{aligned}$$

□